TITLE: Netflix – Data Exploration and Visualization

Context:

Netflix is one of the most popular media and video streaming platforms with over 10000 movies or TV shows. As of mid-2021, they have over 222M Subscribers globally. This dataset consists such as - cast, directors, ratings, release year, duration, etc.

Problem Statement:

- Analyse the Netflix dataset to identify content trends, audience preferences, and growth patterns.
- The objective is to determine which type of content Netflix should focus on and how it can grow its business.

Objective of Analysis:

- Compare Movies and TV Shows
- Analyse content growth over the years
- Study country-wise distribution
- Identify best time to release content
- Analyse actors and ratings
- Provide business insights and recommendations

Analysis:

❖ Load Data

```
import pandas as pd

df = pd. read_csv("netflix.csv")
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

❖ Check Shape (size of data)

```
df.shape
```

```
...  (8807, 12)
```

Insight:

- Dataset contains a large number of rows and columns representing Netflix content.

❖ Check Data Info

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Insight:

- Dataset contains both categorical and numerical columns.
- Some columns have missing values.

➢ Check Data Info

```
df.describe(include = 'all')
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	month_added	year_added
count	8807	8807	8807	6173	7982	8807	8709	8807.000000	8803	8804	8807	8807	8709.000000	8709.000000
unique	8807	2	8807	4528	7692	749	NaN	NaN	17	220	514	8775	NaN	NaN
top	s8807	Movie	Zubaan	Rajiv Chilaka	[David Attenborough]	[United States]	NaN	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...	NaN	NaN
freq	1	6131	1	19	19	2818	NaN	NaN	3207	1793	362	4	NaN	NaN
mean	NaN	NaN	NaN	NaN	NaN	NaN	2019-05-23 01:45:29.452290816	2014.180198	NaN	NaN	NaN	NaN	6.653347	2018.887932
min	NaN	NaN	NaN	NaN	NaN	NaN	2008-01-01 00:00:00	1925.000000	NaN	NaN	NaN	NaN	1.000000	2008.000000
25%	NaN	NaN	NaN	NaN	NaN	NaN	2018-04-20 00:00:00	2013.000000	NaN	NaN	NaN	NaN	4.000000	2018.000000
50%	NaN	NaN	NaN	NaN	NaN	NaN	2019-07-12 00:00:00	2017.000000	NaN	NaN	NaN	NaN	7.000000	2019.000000
75%	NaN	NaN	NaN	NaN	NaN	NaN	2020-08-26 00:00:00	2019.000000	NaN	NaN	NaN	NaN	10.000000	2020.000000
max	NaN	NaN	NaN	NaN	NaN	NaN	2021-09-25 00:00:00	2021.000000	NaN	NaN	NaN	NaN	12.000000	2021.000000
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	NaN	3.431434	1.567961

Insight :

- Release year shows content distribution over time
- Most content is from recent years

➢ Missing Values:

```
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

Insight:

- Director, cast, and country columns have missing values which may effect analysis.

➢ Data Pre-processing

```
df['date_added'] = df['date_added'].astype(str)


df['date_added'] = df['date_added'].str.strip()

df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')

df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month

df['type'] = df['type'].astype('category')
df['rating'] = df['rating'].astype('category')
```

Insight:

- Converted date_added to date time and extracted year_added and month_added for time –based analysis

➢ Non-Graphical Analysis:

```
df[['type', 'rating']].nunique()
```

	0
type	2
rating	17

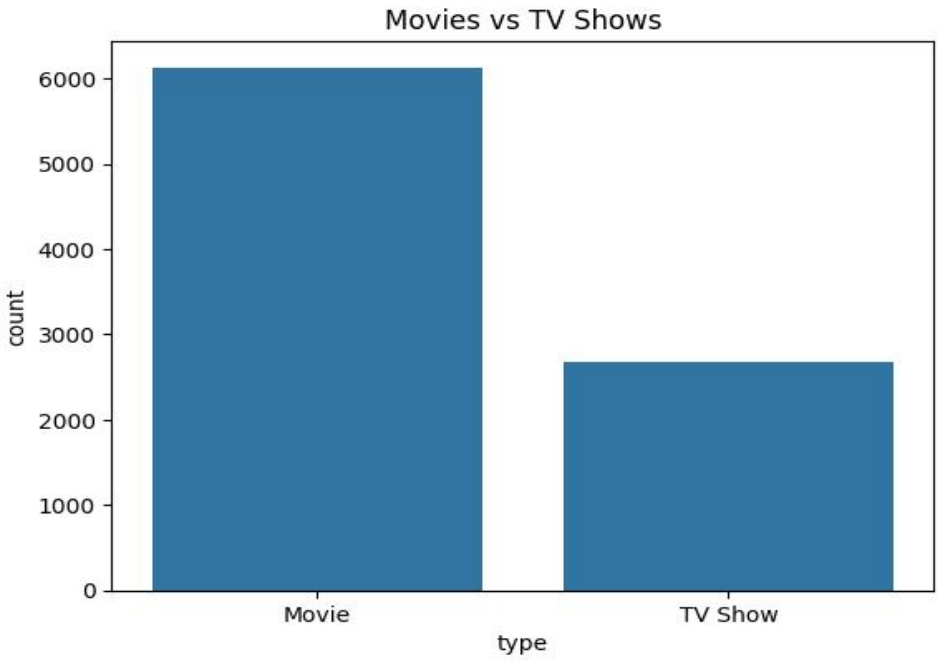
dtype: int64

Insight:

- Few unique types and ratings, showing well-defined content categories.

❖ Movies vs TV Shows

```
import seaborn as sns

import matplotlib.pyplot as plt


sns.countplot(x='type', data=df)
plt.title("Movies vs TV Shows")
plt.show()
```

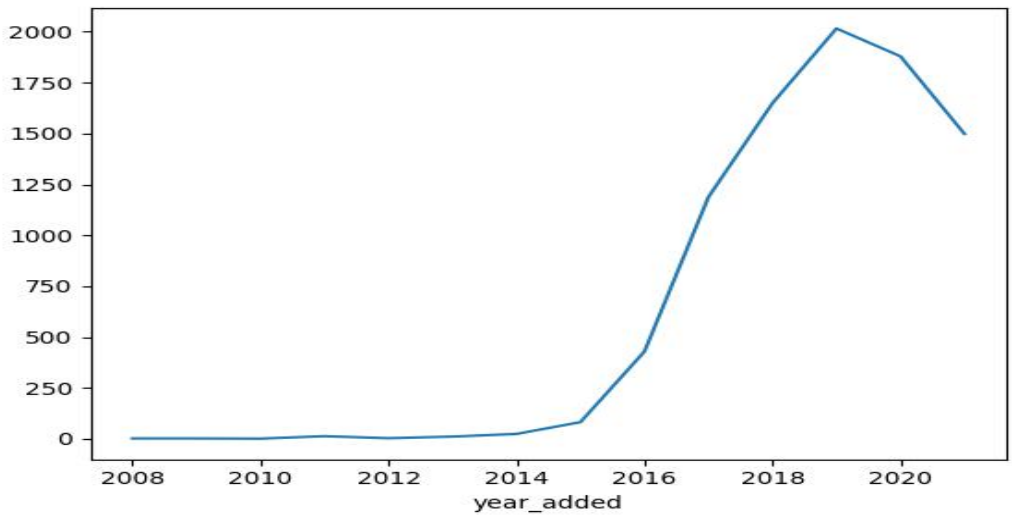


Insight:

- Movies dominate the platform, showing Netflix focuses more on short-duration content for quick audience engagement.

▪ Content Growth Over Years

```
df['year_added'].value_counts().sort_index().plot()
plt.show()
```

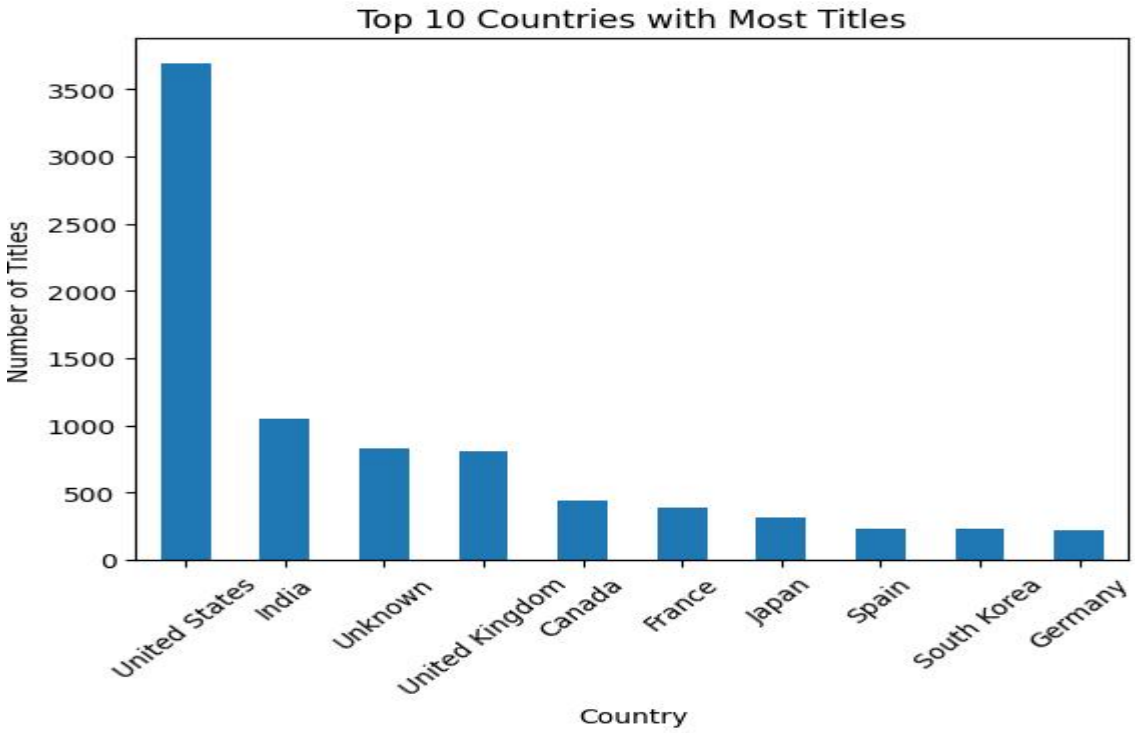


Insight:

- Netflix content increased rapidly after 2015, indicating aggressive expansion to compete with other streaming platforms.

❖ Country Analysis

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("netflix.csv")


df['country'] = df['country'].fillna('Unknown')


df['country'] = df['country'].str.split(',')
df_explode = df.explode('country')


df_explode['country'] = df_explode['country'].str.strip()


top_countries = df_explode['country'].value_counts().head(10)


top_countries. Plot(kind='bar')
plt.xticks(rotation=45)
plt.xlabel('Country')
plt.ylabel('Number of Titles')
plt.title('Top 10 Countries with Most Titles')
plt.show()
```

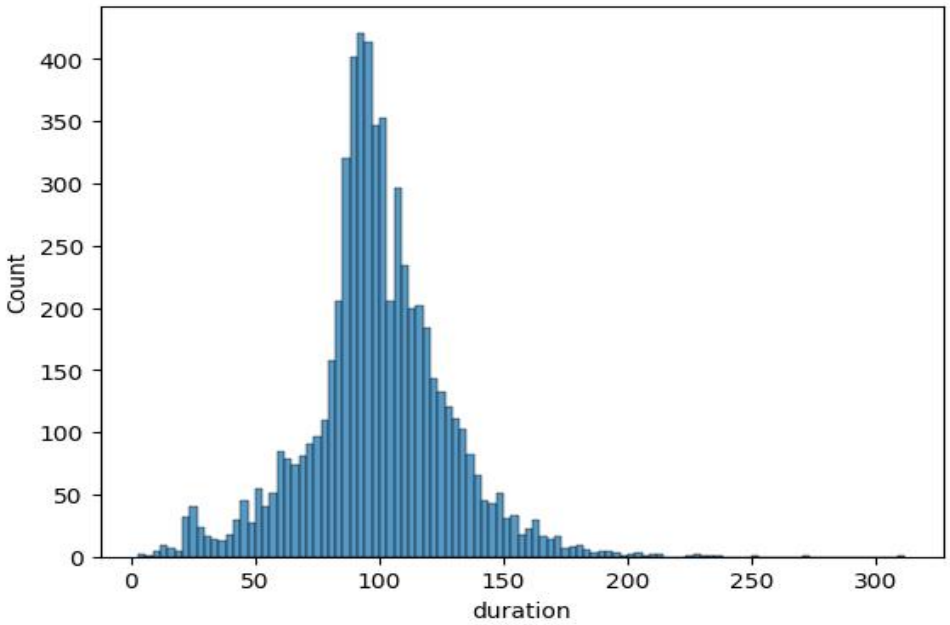


Insight:

- USA and India contribute the most content, highlighting key markets for Netflix's growth and investment.

❖ Movie Duration Analysis

```
movies = df[df['type'] == 'Movie'].copy()
movies['duration'] = movies['duration'].str.replace(' min','')
movies['duration'] = pd.to_numeric(movies['duration'], errors='coerce')

sns.histplot(movies['duration'])
plt.show()
```

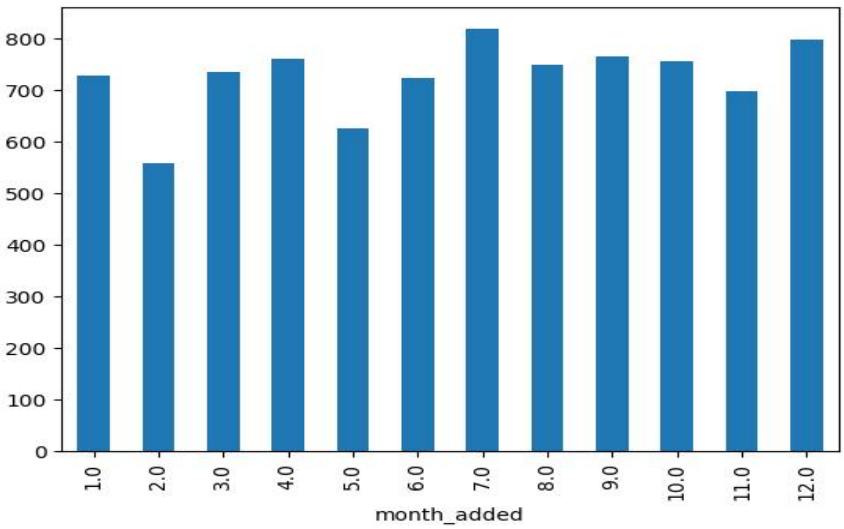


Insight:

- Most movies fall in the 90-120 minutes range, suggesting a standard preference for feature-length films

❖ Best Time to Release

```
df['month_added'].value_counts().sort_index().plot(kind='bar')
plt.show()
```



Insight:

- Certain months show higher content additions, indicating Netflix releases content strategically to maximize audience engagement.

❖ Actor Analysis

```
df['cast'] = df['cast'].str.split(',')
df_cast = df.explode('cast')

df_cast['cast'].value_counts().head(10)
```

...

cast	count
Anupam Kher	39
Rupa Bhimani	31
Takahiro Sakurai	30
Julie Tejwani	28
Om Puri	27
Rajesh Kava	26
Shah Rukh Khan	26
Andrea Libman	25
Paresh Rawal	25
Boman Irani	25

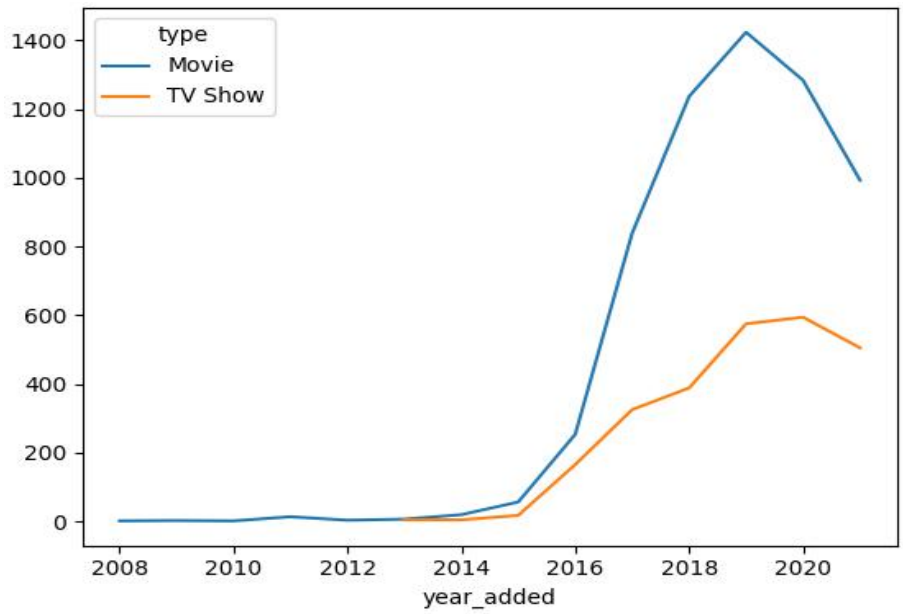
dtype: int64

Insight:

- Some actors appear frequently, suggesting Netflix collaborates with popular actors to attract more viewers.

➢ Trend Shift (Movies vs TV Shows over time)

```
df['date_added'] = pd.to_datetime(df['date_added'],errors='coerce')
df['year_added'] = df['date_added'].dt.year

df.groupby(['year_added','type']).size().unstack().plot()

plt.show()
```

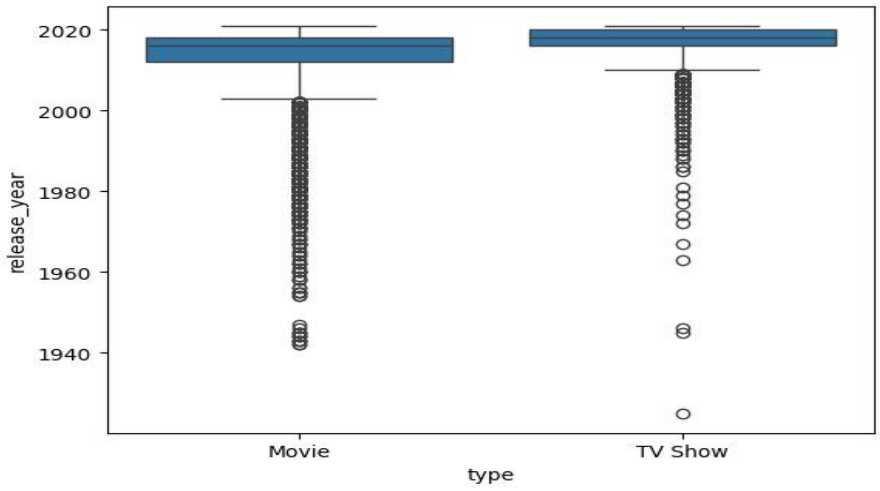


Insight:

- While movies dominate overall , TV shows have increased in recent years, indicating a shift towards long-term user engagement.

❖ Boxplot Analysis

```
sns.boxplot(x='type', y='release_year', data=df)
plt.show()
```



Insight:

- TV shows are generally more recent than movies, showing Netflix is focusing more or newer series content

Missing Values & Outlier Observation:

Insight:

Missing values are present in key columns. Some extreme values exist but most data is concentrated in recent years.

Final Business Insights:

- Netflix focuses more on Movies
- Content increased significantly after 2015
- USA and India are major contributors
- TV Shows are increasing over time

Recommendations:

- Produce more Movies in popular duration range
- Invest in high-performing countries like India and USA
- Focus on content types that engage users longer
- Release content strategically in peak months